
ShopnilOS: A GNU/Linux Based Distribution with Custom System Administration and Process Monitoring Tools

CIT 322 :: Operating System Sessional

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1. Introduction

This project demonstrates the practical application of Operating System principles through the development of a custom, lightweight Linux distribution named ShopnilOS, alongside a suite of system administration and process monitoring tools. ShopnilOS is built by customizing the ArchLinux ISO to create a bare-minimum live environment populated only with essential modern applications. It utilizes archinstall for a streamlined, script-free base installation, ensuring a lightweight footprint. The environment is graphical-ready, customized with the Hyprland Wayland compositor, SDDM, Waybar, Rofi, and Waypaper.

To bridge the gap between low-level system configuration and user-space process management, complementary tools were developed: Archmate for comprehensive system administration, a Chaotic AUR installer script, and Tree_tasker, a Python and PySide6-based graphical task manager with web integration for advanced process hierarchy visualization.

2. Objectives

- To understand Linux system architecture by customizing and building a bootable Arch-based ISO.
- To automate core OS maintenance tasks including package, user, repository, and configuration management.
- To visualize and monitor system resource allocation (CPU, RAM, Disk, Network) per application in real-time.
- To map and interactively explore parent-child process relationships (PIDs) within the operating system.

3. Problem Statement

Modern operating systems often come pre-packaged with heavy, unnecessary background services (bloatware) that consume valuable hardware resources. Furthermore, traditional command-line process monitors (like top or ps) provide dense tabular data that makes it difficult for users to intuitively understand parent-child process relationships and exact resource distribution. There is a need for a streamlined, customizable OS base paired with an intuitive, visual process manager that clearly maps process hierarchies and resource bottlenecks.

4. Related Work

1. General-purpose Linux distributions (e.g., Ubuntu, Linux Mint) provide functional environments but lack the minimalism required for strict resource control. On the monitoring side, tools like htop, gnome-system-monitor, or Windows Task Manager offer system overviews but often lack strict tree-based PID mapping paired with detailed pie-chart breakdowns for CPU, RAM, Disk, and Network per application. This project combines the minimalism of Arch Linux with a custom-built, highly visual process monitoring suite.

5. Scope

The scope of this project is currently limited to x86_64 system architectures. The distribution, ShopnilOS, is tailored specifically for Arch-based package management (pacman, AUR). The Tree_tasker application is designed to fetch and visualize process data strictly for the local host machine, though its web interface lays the groundwork for future remote monitoring capabilities.

6. Methodology

6.1. Technology Stack

1. OS Customization: Archiso builder, archinstall.
2. Window Management / UI: Hyprland (Wayland), SDDM, Waybar, Rofi, Waypaper.
3. Scripting & Administration: Bash scripting (Archmate, Chaotic AUR Installer).
4. Application Development: Python, PySide6 (GUI framework for Tree_tasker).
5. Web Technologies: HTML, CSS, JavaScript (for the interactive process web interface).

6.2. Design Principles

Minimalism (KISS): Keep the core OS live ISO as small as possible, relying on native tools like archinstall rather than custom, hard-to-maintain installation scripts.

Modularity: Separate system administration (Archmate) from real-time process monitoring (Tree_tasker) to ensure system stability.

Visual Clarity: Use tree structures and pie charts to translate raw OS process data into easily digestible visual information.

7. Future Plans

Future iterations of this project aim to implement remote system administration directly into Archmate, allowing configuration over SSH. For Tree_tasker, plans include adding the capability to terminate or pause specific processes directly from the interactive web interface, and expanding the data polling to support distributed containerized environments (like Docker metrics).

8. Result

The project resulted in a fully functional, bootable custom Linux ISO (ShopnilOS) that successfully integrates a modern Wayland environment on minimal base packages. The administration scripts (Archmate and Chaotic AUR Installer) successfully automate tedious

system updates, cleanups, and user management. Finally, the Tree_tasker application accurately polls kernel data to display real-time parent-child process trees and accurately renders CPU, RAM, Disk, and Network usage into interactive pie charts across both desktop and web interfaces.

THE END